Landau-Gauge $W^{\pm}$ Propagator from the b-Channel: Gauge Fixing, Gribov Copies, and the Cross-Projected Correlator in $\mathrm{SU}(2)$

T. Firestone

Abstract

Paper 5 established the exact algebraic decomposition $G_{\text{diag}}(t) + G_{\text{cross}}(t) = G_{\text{full}}(t)$ in the $\mathrm{Cl}(0,2) \cong \mathbb{H}$ framework and identified $G_{\text{cross}}(t) = 8\sum_{x,\mu} \mathrm{Re}[b_t \bar{b}_0]$ as the b-channel bilinear carrying the charged ($W^{\pm}$) content of the embedded SU(2) link. That identification was, however, gauge dependent: G_{cross} is not gauge invariant, no gauge prescription had been imposed, and no propagator had been extracted. This paper closes the gap. We implement Landau gauge fixing on the lattice through steepest-descent and Fourier-accelerated algorithms, prove a convergence-rate lemma (Lemma 4), and characterise the Gribov copy systematic through a multi-start best-copy strategy and an explicit variation bound (Proposition 7). The central result, Theorem 2, promotes the formal perturbative identification of Paper 5 to a fully specified, error-budgeted quantity: the gauge-fixed correlator $G^{\text{gf}}_{\text{cross}}(t)$ yields the zero-spatial-momentum $W^{\pm}$ temporal propagator $D_{W^{\pm}}(t)$ with a complete three-term error budget. Corollary 6 characterises the sign structure of the ensemble-averaged correlator and defines a coupling threshold β_c for the onset of non-perturbative sign violation, which is studied empirically through an extended ensemble programme spanning $N \in \{6, 8, 12, 16\}$ and $\beta_{\text{eff}} \in \{2.0, \ldots, 20.0\}$ with a $1/N^2$ finite-volume extrapolation. We establish SU(2) colour symmetry between the b-channel propagator and the full Landau-gauge gluon propagator, enabling comparison to the SU(2) reference dataset of Cucchieri–Mendes (with Bogolubsky et al. providing a qualitative SU(3) confirmation of the same decoupling behaviour). Finally, the fermionic sector is set up: we define the cross-projected propagator $S_{\text{cross}} = P^{\text{EW}}_{+} S\, P^{\text{EW}}_{-}$, prove that it vanishes identically for an SU(2) singlet (Lemma 5), and specify the weak-coupling verification (checks D28/D29) that the doublet cross-projected propagator is non-zero. Throughout, the embedding does not by itself distinguish $\mathrm{SU}(2)_L$ from $\mathrm{SU}(2)_R$; this requires external fermionic input, the full non-perturbative test of which is deferred to Paper 7.

1 Introduction

Paper 5 of this series established, within the $\mathrm{Cl}(0,2) \cong \mathbb{H}$ framework, the exact identity

$$G_{\text{diag}}(t) + G_{\text{cross}}(t) = G_{\text{full}}(t), \tag{1}$$

(Theorem 1, Paper 5) together with the Two-Observable Decomposition (Corollary 5, Paper 5), in which the cross-channel correlator

$$G_{\text{cross}}(t) = 8\sum_{x,\mu} \mathrm{Re}\big[b_t(x,\mu)\, \bar{b}_0(x,\mu)\big] \tag{2}$$

is identified as the b-channel bilinear, with $b = (U)_{01}$ the off-diagonal element of the SU(2) link $U = \left(\begin{smallmatrix} a & b \\ -\bar{b} & \bar{a} \end{smallmatrix}\right)$, $|a|^2 + |b|^2 = 1$. At weak coupling $b = (ig/\sqrt{2})W^{+}_{\mu} + O(g^3)$, so G_{cross} carries the charged ($W^{\pm}$) content of the embedded gauge field, while the diagonal correlator G_{diag} carries the neutral W^3/A^3 content (Paper 5, §7.3).

The inherited gap. The Paper 5 identification was incomplete in six specific respects. (i) G_{cross} is not gauge invariant (gauge non-invariance, Paper 5 §8.1), yet no gauge prescription had been fixed. (ii) The formal identification of Paper 5 §8.3 applied to spatial links only, because Papers 4–5 worked in temporal gauge $U_0(x) = \mathbb{I}_2$. (iii) No control of Gribov copy ambiguity had been established. (iv) The $W^{\pm}$ temporal propagator $D_{W^{\pm}}(t)$ had not been extracted from any ensemble. (v) No error budget existed. (vi) No comparison to independent lattice data had been made. Paper 6 addresses all six, organising its verification checks D19–D29 against them.

What this paper establishes. The centrepiece is Theorem 2 (Section 4), which promotes the Paper 5 §8.3 formal identification to a fully specified, error-budgeted quantity. The route requires: a Landau gauge functional summed over all four link directions, lifting the temporal-link restriction of Papers 4–5 (Section 2); a convergence-rate result for the gauge-fixing functional (Lemma 4); a multi-start best-copy strategy with an explicit Gribov variation bound (Proposition 7, Section 3); and an extended ensemble programme with finite-volume extrapolation (Section 5). Corollary 6 characterises the sign structure of $G^{\mathrm{gf}}_{\mathrm{cross}}$, and Section 7 delivers its empirical content. Colour symmetry between the b-channel and the full gluon propagator (Section 6) enables comparison to the SU(2) reference dataset of Cucchieri–Mendes [3] (Bogolubsky et al. [4] confirm the same decoupling in SU(3)). The fermionic groundwork (Section 8) defines the cross-projected propagator and proves the singlet vanishing (Lemma 5).

Scope limits. The following hard scope limits are carried forward from Papers 4–5 and remain in force except where explicitly addressed: (SL1) no $\mathrm{SU}(2)_L/\mathrm{SU}(2)_R$ distinction without fermion content; (SL2) no chiral fermion claims without a Nielsen–Ninomiya-bypass action; (SL3) no W/Z mass ratio, Weinberg angle, or $U(1)_Y$ mixing; (SL4) no Higgs mechanism; (SL5) no non-perturbative $G_{\mathrm{cross}} \leftrightarrow W^{\pm}$ identification without the Landau-gauge apparatus. SL5 is discharged in Section 3; SL1 and SL2 are addressed (not fully retired) in Section 8.

> *L/R caveat (verbatim, first occurrence): The embedding does not, by itself, distinguish* $\mathrm{SU}(2)_L$ *from* $\mathrm{SU}(2)_R$*; this requires external fermionic input.*

Infrastructure. The simulation functions implemented in this paper are, in paper-local numbering, P6-F1 (steepest descent) and P6-F2 (Fourier acceleration) in Section 2; P6-F3 (best-copy strategy) in Section 3; P6-F4 (gauge-fixed correlator) in Section 4; P6-F6 through P6-F9 (sign analysis) in Section 7; and P6-F10 through P6-F13 (fermionic sector) in Section 8. New numbered results are Lemma 4 (Section 2), Proposition 7 (Section 3), Theorem 2 and Corollary 6 (Section 4), and Lemma 5 (Section 8).

2 Landau Gauge Fixing: Algorithm and Implementation

The algebraic programme of Papers 4–5 established G_{cross} exactly as a b-channel observable but left it gauge dependent (gauge non-invariance, Paper 5 §8.1). Before a propagator can be extracted, a gauge prescription must be fixed. This section develops the Landau gauge-fixing machinery: the functional F_L is defined (§2.1), the lattice Landau condition and convergence criterion are derived (§2.2), the steepest-descent algorithm is specified and its monotone-increase property proved (§2.3), the Fourier-accelerated algorithm is specified (§2.4), and Lemma 4 establishes convergence rates and the regime of preference (§2.5). No measurement of $G^{\mathrm{gf}}_{\mathrm{cross}}$ is performed here; that is Section 4. The multiplicity of local maxima of F_L (Gribov copies) is treated in Section 3.

2.1 The Landau Gauge Functional

Definition 1 (Landau Gauge Functional)**.** For a set of SU(2) gauge transformations $\{g(x)\}$ on the lattice $\Lambda = \{0, \dots, N-1\}^3 \times \{0, \dots, N_t - 1\}$ with periodic boundary conditions, the Landau gauge functional is

$$F_L[\{g(x)\}] = \mathrm{Re} \sum_{x,\mu} \mathrm{Tr}\big[U_\mu^g(x)\big], \tag{3}$$

where $U_\mu^g(x) = g(x)\, U_\mu(x)\, g^\dagger(x+\hat{\mu})$ and the sum runs over all links: $x \in \Lambda$ and $\mu \in \{0, 1, 2, 3\}$, including the temporal direction $\mu = 0$.

Since $\mathrm{Tr}[U_\mu^g] + \mathrm{Tr}[(U_\mu^g)^\dagger] = 2\,\mathrm{Re}\,\mathrm{Tr}[U_\mu^g]$, the functional is real. Each link contributes at most $\mathrm{Re}\,\mathrm{Tr}[\mathbb{I}_2] = 2$ and at least -2, giving the uniform bounds

$$0 \le F_L[\{g(x)\}] \le F_L^{\max} := 2\,dV, \tag{4}$$

with $d = 4$ and $V = N^4$, so $F_L^{\max} = 8V$.

Remark. Papers 4–5 imposed temporal gauge $U_0(x) = \mathbb{I}_2$ before measurement, restricting F_L to spatial links (maximum $6V$). Paper 5 §8.3 noted this as a caveat. Paper 6 lifts the restriction: the temporal links are live variables and (3) sums over all four directions, matching the standard prescription of [3, 4] and enabling the quantitative comparison of Section 6.

Landau gauge is the choice of representatives $\{g^*(x)\}$ globally maximising F_L,

$$\{g^*(x)\} = \underset{\{g(x)\} \in \mathrm{SU}(2)^V}{\mathrm{argmax}} \; F_L[\{g(x)\}]. \tag{5}$$

Global maximisation is generically obstructed by multiple local maxima on each gauge orbit — the Gribov copies [6, 7] — treated in Section 3.

2.2 The Lattice Landau Condition

Definition 2 (Lattice Gauge Field)**.** For a gauge-transformed link, the lattice gauge field is

$$A_\mu^a(x) = \frac{1}{2i}\, \mathrm{Tr}\big[T^a \big(U_\mu^g(x) - (U_\mu^g(x))^\dagger\big)\big], \tag{6}$$

with $T^a = \sigma^a/2$ the SU(2) generators.

Definition 3 (Lattice Divergence and Convergence Measure)**.** The lattice divergence and the global convergence measure are

$$(\partial_\mu A_\mu)^a(x) = \sum_{\mu=0}^{3} \big[A_\mu^a(x) - A_\mu^a(x-\hat{\mu})\big], \tag{7}$$

$$\Theta = \frac{1}{d\, N_c\, V} \sum_x \sum_{a=1}^{3} \big[(\partial_\mu A_\mu)^a(x)\big]^2, \tag{8}$$

with $d = 4$, $N_c = 2$. The factor $d = 4$ in the denominator is essential; using $d = 3$ would inflate Θ by 33% and cause premature termination.

The lattice Landau condition is $\Theta = 0$; the numerical criterion adopted throughout is

$$\Theta < \varepsilon_{\mathrm{gauge}} = 10^{-14}. \tag{9}$$

Under $g(x) \to (\mathbb{I} + i\varepsilon\, \omega^a(x) T^a) g(x)$,

$$\delta F_L = -2\varepsilon \sum_{x,a} \omega^a(x)\, (\partial_\mu A_\mu)^a(x) + O(\varepsilon^2), \tag{10}$$

so stationarity requires $(\partial_\mu A_\mu)^a(x) = 0$ for all x, a — the lattice Landau condition.

2.3 Steepest-Descent Gauge Fixing (P6-F1)

Definition 4 (Local Staple)**.** With all $g(y)$, $y \neq x$, fixed, the site-x contribution to F_L is $f_x[g(x)] = \mathrm{Re}\,\mathrm{Tr}[g(x)W(x)]$ with staple

$$W(x) = \sum_{\mu=0}^{3} [U_\mu^g(x) + (U_\mu^g(x-\hat{\mu}))^\dagger], \tag{11}$$

parametrised as $W(x) = w_0\mathbb{I}_2 + i\,\boldsymbol{w}\cdot\boldsymbol{\sigma}$ with $|W(x)| = (w_0^2 + |\boldsymbol{w}|^2)^{1/2}$.

Proposition 1 (Local Maximisation on SU(2))**.** *For fixed $W(x)$ with $|W(x)| > 0$, the unique maximiser of* $\mathrm{Re}\,\mathrm{Tr}[g(x)W(x)]$ *over* $g(x) \in \mathrm{SU}(2)$ *is* $g^*(x) = W^\dagger(x)/|W(x)|$*.*

Proof. Write $g = g_0\mathbb{I}_2 + i\boldsymbol{g}\cdot\boldsymbol{\sigma}$, $g_0^2 + |\boldsymbol{g}|^2 = 1$. Using $\mathrm{Tr}[\mathbb{I}_2] = 2$ and $\mathrm{Tr}[\sigma^a\sigma^b] = 2\delta^{ab}$,

$$\mathrm{Re}\,\mathrm{Tr}[g(x)W(x)] = 2(g_0w_0 - \boldsymbol{g}\cdot\boldsymbol{w}) = 2\,(g_0, \boldsymbol{g})\cdot(w_0, -\boldsymbol{w}).$$

By Cauchy–Schwarz with $|(g_0, \boldsymbol{g})| = 1$, this is at most $2|W(x)|$, with equality iff $(g_0, \boldsymbol{g}) \propto (w_0, -\boldsymbol{w})$, i.e. $g^*(x) = W^\dagger(x)/|W(x)|$; $\det g^* = 1$ so $g^* \in \mathrm{SU}(2)$. □

Algorithm P6-F1 (steepest descent). Initialise $g^{(0)}(x) = R(x)$, Haar-random SU(2) per site. Iterate: for each site x compute $W^{(k)}(x)$ and set $g^{(k+1)}(x) = (W^{(k)}(x))^\dagger/|W^{(k)}(x)|$, updating the adjacent links; after each full sweep compute Θ; stop when $\Theta < \varepsilon_{\mathrm{gauge}}$, recording the sweep count k^*; flag non-convergence beyond $k_{\max} = 10\,000$. Monotone increase of F_L is Lemma 4(i). An over-relaxation variant ($\omega \approx 1.7$–1.9) reduces the sweep count by $O(N)$ but is not used for the $N = 6$ ensembles.

2.4 Fourier-Accelerated Gauge Fixing (P6-F2)

Steepest descent converges slowly for long-wavelength modes ($\rho_{\mathrm{SD}} \approx 1 - \pi^2/N^2$). The Fourier-accelerated (FA) algorithm preconditions the update in momentum space, reducing the sweep count to $O(N)$ while preserving monotone increase in the linearised regime; it is standard for large volumes [3].

Algorithm P6-F2. Each pass: compute $A_\mu^a(x)$, the divergence (7), and Θ; stop if $\Theta < \varepsilon_{\mathrm{gauge}}$. Otherwise Fourier transform the divergence,

$$\tilde{p}^a(p) = \sum_x e^{2\pi i\,p\cdot x/N}(\partial_\mu A_\mu)^a(x), \qquad \tilde{p}^a(0) := 0, \tag{12}$$

precondition with the lattice Laplacian eigenvalue $f(p) = \sum_\mu 4\sin^2(\pi p_\mu/N)$,

$$\delta\tilde{\omega}^a(p) = \tilde{p}^a(p)/f(p), \tag{13}$$

invert the transform, exponentiate $h(x) = \exp(i\alpha\,\delta\omega^a(x)T^a)$ with step size α, and update $g^{(k+1)}(x) = h(x)g^{(k)}(x)$. The step size is tuned per β_{eff} ($\alpha = 0.08$ at $\beta_{\mathrm{eff}} \geq 9$ down to $\alpha = 0.20$ at $\beta_{\mathrm{eff}} \leq 3$), with optimal value $\alpha_{\mathrm{opt}} = 2/(f_{\min} + f_{\max})$, $f_{\max} = 16$. The zero-mode projection removes the global rotation. `scipy.fft` (1.16.3) implements the transforms.

2.5 Lemma 4: Convergence Rate of F_L

Lemma 4 (Convergence Rate of F_L on SU(2) Ensembles)**.** *Let Λ be hypercubic with N sites per direction, $N_t = N$, $V = N^4$; $\{U_\mu(x)\}$ a thermalised* SU(2) *ensemble at coupling β_{eff}; $\varepsilon_{\mathrm{gauge}} = 10^{-14}$.*

(i) ***Monotone increase.*** *Every site update in P6-F1 satisfies* $F_L[\{g^{(k+1)}\}] \geq F_L[\{g^{(k)}\}]$*, with equality iff* $\Theta = 0$ *everywhere. The algorithm converges in finite* k^* *to a local maximum of* F_L*.*

(ii) β_{eff} ***dependence.*** $N_{\text{iter}}^{\text{SD}} \approx (N^2/\pi^2)\,(1+O(8/\beta_{\text{eff}}))\,|\log \varepsilon_{\text{gauge}}| = O\big(V^{1/2}(1+c/\beta_{\text{eff}})|\log \varepsilon_{\text{gauge}}|\big)$*.*

(iii) ***Fourier acceleration.*** $N_{\text{iter}}^{\text{FA}} = O\big(V^{1/4}(1+c/\beta_{\text{eff}})|\log \varepsilon_{\text{gauge}}|\big)$ *passes, each of cost* $O(V \log V)$*. The speedup over SD is* $O(N/\log N)$*, exceeding unity for* $N \geq 12$*.*

Proof. (i) Fixing $g(y)$, $y \neq x$, the update sets $\mathrm{Re}\,\mathrm{Tr}[g^{(k+1)}(x)W(x)] = 2|W(x)|$, the global maximum, while $\mathrm{Re}\,\mathrm{Tr}[g^{(k)}(x)W(x)] \leq 2|W(x)|$; hence $\Delta F_x \geq 0$ at every site and $F_L^{(k+1)} \geq F_L^{(k)}$. Equality requires $W(x) \propto g^{(k)}(x)^\dagger$ for all x, the stationarity condition $\Theta = 0$. Bounded above by $8V$ and non-decreasing, the sequence converges to a local maximum.

(ii) Near a local maximum, the second variation of F_L is governed by the Faddeev–Popov operator M_{FP}, which in the $\beta_{\text{eff}} \to \infty$ limit reduces to the lattice Laplacian with eigenvalues $\lambda_p = \sum_\mu 4\sin^2(\pi p_\mu/N)$, smallest non-zero $\lambda_1 \approx 4\pi^2/N^2$, largest $\lambda_{\max} = 16$, condition number $\kappa \approx (4/\pi^2)N^2$. The SD reduction factor per sweep is $\rho_{\text{SD}} \approx 1 - \pi^2/N^2$, giving $N_{\text{iter}}^{\text{SD}} \approx (N^2/\pi^2)|\log(\varepsilon_{\text{gauge}}/\Theta_0)|$, with the $O(g^2)$ correction at finite β_{eff} multiplying by $(1 + O(8/\beta_{\text{eff}}))$. At strong coupling ($\beta_{\text{eff}} \lesssim 3$) near-zero Faddeev–Popov modes signal Gribov-horizon proximity; this is a physical effect, flagged in D19.

(iii) FA inverts the Laplacian part of M_{FP} each pass, so the spectral *gap* square-roots, $(1 - \rho_{\text{FA}}) \approx (1 - \rho_{\text{SD}})^{1/2}$; with $1 - \rho_{\text{SD}} \approx \pi^2/N^2$ this gives $1 - \rho_{\text{FA}} \approx \pi/N$ and $O(N)$ passes. SD costs $O(V^{3/2})$ total operations, FA costs $O(V^{5/4}\log V)$; the speedup is $O(V^{1/4}/\log V) = O(N/\log N)$, giving heuristically (hidden constants set to unity) $S(12) \approx 1.20$, $S(16) \approx 1.44$. □

Checks D19–D21. D19 verifies $\Theta < 10^{-14}$ after every gauge-fixing call (expected pass rate $\geq 95\%$ at $\beta_{\text{eff}} \in \{5, \ldots, 12\}$; lower at strong coupling with stalling flagged, not failed). D20 verifies the per-site condition $\max_{x,a} |(\partial_\mu A_\mu)^a(x)| < \varepsilon_{\text{site}} = \sqrt{8V \cdot 10^{-14}}$. D21 verifies (a) $F_L^{\text{gf}} > F_L^{\text{initial}}$, (b) $F_L^{\text{gf}}/(8V) \in [R_{\min}(\beta_{\text{eff}}), 1]$, and (c) extensive volume scaling within 5%.

2.6 Algorithm Comparison and Regime Decision

The re-processed Paper 4/5 ensembles ($N = 6$, $\beta_{\text{eff}} \in \{5, 6, 9, 12\}$) are gauge-fixed with P6-F1; the marginal FA speedup at $N = 6$ does not justify the FFT infrastructure. All extended ensembles ($N \in \{8, 12, 16\}$, $\beta_{\text{eff}} \in \{2, \ldots, 20\}$) use P6-F2. In both cases $\Theta < 10^{-14}$ with $d = 4$ in (8), and the best-copy strategy P6-F3 (Section 3) with ≥ 5 random starts precedes any measurement.

3 Gribov Copy Analysis and Systematic Uncertainty

A local maximum of F_L need not be unique on a gauge orbit. The existence of multiple local maxima — Gribov copies [6, 7] — means P6-F1/F2 may select different representatives on different runs, introducing an irreducible systematic. This section quantifies that uncertainty for G_{cross}.

3.1 The Gribov Problem in SU(2) Lattice Landau Gauge

Stationarity of F_L ($\Theta = 0$) is necessary but not sufficient for uniqueness. Gribov [6] showed multiple stationary points exist on every orbit; Singer [8] proved a topological no-go theorem for global gauge conditions over a compact base. The first Gribov region Ω consists of Landau-gauge configurations with positive-definite Faddeev–Popov operator; the fundamental modular region $\Lambda \subset \Omega$ of global maxima underlies the Gribov–Zwanziger programme [7], beyond our scope. In practice one adopts either first-copy or best-copy prescriptions; [3] quantifies the difference for the SU(2) gluon propagator (and [4] for SU(3)).

Because $G_{\rm cross}$ is not gauge invariant (gauge non-invariance, Paper 5 §8.1), distinct copies give distinct values $G^{(k)}_{\rm cross}(t) = 8\sum_{x,\mu} \mathrm{Re}[b^{(k)}_t \bar{b}^{(k)}_0]$. This copy-dependence must be characterised before $G^{\rm gf}_{\rm cross}$ can serve as a physical observable. SL5 was conditionally lifted by Landau-gauge convergence (D19–D21); its full discharge additionally requires the Gribov systematic to be bounded, which Proposition 7 provides.

3.2 Best-Copy Strategy P6-F3

Algorithm P6-F3. For each configuration: (1) generate $n_{\rm start} \geq 5$ independent Haar-random gauge transformations $\{R^{(k)}(x)\}$; (2) apply P6-F1 ($N = 6$) or P6-F2 ($N \geq 8$) from each start, reaching $\Theta^{(k)} < \varepsilon_{\rm gauge}$; (3) record $F^{(k)}_L$ and $G^{(k)}_{\rm cross}(t)$; (4) select the best copy $k^* = \mathrm{argmax}_k\, F^{(k)}_L$; (5) use $\{U^{\rm best}\}$ for all subsequent measurements. We write $G^{\rm best}_{\rm cross}$, $G^{\rm first}_{\rm cross}$, $\Delta G_{\rm cross}(t) = \max_k G^{(k)}_{\rm cross}(t) - \min_k G^{(k)}_{\rm cross}(t)$, and $\Delta_{\rm rel}(t) = \Delta G_{\rm cross}(t)/|G^{\rm best}_{\rm cross}(t)|$.

3.3 Proposition 7: Bounds on Gribov Copy Variation

Proposition 7 (Bounds on Gribov Copy Variation of $G_{\rm cross}$)**.** *Let $\{U^{(k)}_\mu(x)\}$, $k = 1, \ldots, n_{\rm start} \geq 5$, be Gribov copies of the same configuration produced by P6-F3, all satisfying the Landau condition to $\varepsilon_{\rm gauge} = 10^{-14}$.*

(i) ***Algebraic variation bound.*** *For any two copies k, l,*

$$|G^{(k)}_{\rm cross}(t) - G^{(l)}_{\rm cross}(t)| \leq 8dV(\delta_t + \delta_0), \tag{14}$$

with $\delta_t = \max_{x,\mu} |b^{(k)}_t - b^{(l)}_t|$, $\delta_0 = \max_{x,\mu} |b^{(k)}_0 - b^{(l)}_0|$. The bound is not saturated on thermalised ensembles; the relative variation satisfies $\Delta_{\rm rel} = O(1)$ in the coupling.

(ii) ***Best-copy improvement.*** *Since $G_{\rm cross}$ is not gauge invariant, $G^{\rm best}_{\rm cross} \neq G^{\rm first}_{\rm cross}$ generically; best-copy selection reduces fluctuations (driving links toward $\mathbb{I}_2$, $b = 0$) but imposes no definite ordering on $G_{\rm cross}$.*

(iii) ***$\beta_{\rm eff}$-dependent threshold (empirical, D22).*** *Three regimes: (a) $\beta_{\rm eff} \geq 9.0$, $N = 6$: $\Delta_{\rm rel} < 1\%$ (manageable); (b) $\beta_{\rm eff} \in \{5, 6\}$, $N = 6$: $\Delta_{\rm rel} \in [1\%, 10\%]$ (moderate); (c) $\beta_{\rm eff} \leq 3.5$, extended ensembles: $\Delta_{\rm rel} > 10\%$ (severe).*

Proof. (i) By the triangle inequality and adding/subtracting $b^{(l)}_t \bar{b}^{(k)}_0$, $|b^{(k)}_t \bar{b}^{(k)}_0 - b^{(l)}_t \bar{b}^{(l)}_0| \leq |b^{(k)}_t - b^{(l)}_t||\bar{b}^{(k)}_0| + |b^{(l)}_t||\bar{b}^{(k)}_0 - \bar{b}^{(l)}_0|$. Since $|b| \leq 1$ by unitarity, summing over dV links gives (14); equality would require $|b| = 1$ everywhere, excluded by Landau gauge. By the b-transformation lemma (Paper 5, §8.1), at weak coupling copies differ by near-identity transformations, so $\delta_t = O(g^2)$ and $|G^{(k)}_{\rm cross} - G^{(l)}_{\rm cross}| = O(g^2 \cdot 4V)$. Then $\Delta_{\rm rel} = O(g^2)/(g^2/2) = O(1)$: the $O(g^2)$ factors cancel, leaving the variation parametrically unsuppressed and controllable only empirically.

(ii) By the gauge non-invariance result (Paper 5 §8.1), distinct gauge-fixed configurations of one orbit give distinct values; $g^{\rm best}$ and $g^{\rm first}$ differ (D22), so the values differ generically. Maximising F_L minimises the average $|b|^2$ per link, reducing sensitivity, but fixes no sign of the difference.

(iii) Empirical; confirmed by D22. The percentage thresholds are order-of-magnitude expectations replaced by measured values in the verification summary. □

SL5 discharge. With Proposition 7 established and D22–D23 passed, SL5 is fully discharged: the non-perturbative identification $G^{\rm gf}_{\rm cross} \leftrightarrow W^{\pm}$ (Theorem 2) rests on verified Landau convergence (D19–D21) and a characterised Gribov systematic.

3.4 Statistics and Reporting Conventions

For each ensemble and timeslice we report $\langle G^{\text{best}}_{\text{cross}}\rangle(t)$, the statistical error $\sigma_{\text{stat}}(t)$, the mean copy range $\langle \Delta G_{\text{cross}}\rangle(t)$, its spread $\sigma_{\text{Gribov}}(t)$, and $\Delta_{\text{rel}}(t)$. The ratio $\sigma_{\text{Gribov}}/\sigma_{\text{stat}}$ dictates reporting: if < 0.1, statistical error only (Gribov controlled); if $\in [0.1, 1]$, both reported; if > 1, Gribov dominant and the ensemble is flagged. The first-copy value $G^{\text{first}}_{\text{cross}}$ is recorded alongside for like-for-like comparison with the first-copy SU(2) data of [3].

Checks D22 and D23. D22 (inverted) requires measurably different copies: $\max_{t,\text{cfg}} \Delta G_{\text{cross}}(t) > \varepsilon_{\text{copy}} = 10^{-6}$. The inversion is motivated by the expectation that thermalised non-Abelian configurations generically admit multiple Gribov copies [6]; Singer [8] establishes the continuum obstruction to a global gauge condition but does not by itself guarantee copy multiplicity for a given finite-lattice configuration. Persistent degeneracy across independent random starts is therefore flagged for a direct check of start independence (RNG and initialisation), not assumed to be a code defect; in the perturbative limit $\beta_{\text{eff}} = 20.0$ near-degeneracy is physical ("copy-degenerate regime"). D23 verifies best-copy stability under a small random perturbation and re-fixing, with failure rates above 5% signalling Gribov-horizon proximity.

4 The Gauge-Fixed $W^{\pm}$ Correlator: Theorem 2

With the ensemble in a controlled gauge (D19–D20) and Gribov ambiguity handled by P6-F3 (D23), we define the gauge-fixed correlator, derive its weak-coupling relation to the $W^{\pm}$ field, and establish Theorem 2 with a three-term error budget. The scope of this section is the Paper 4/5 ensembles ($N = 6$, $\beta_{\text{eff}} \in \{5, 6, 9, 12\}$); extended ensembles are introduced in Section 5.

4.1 The Gauge-Fixed Observable (P6-F4)

Definition 5 (Gauge-Fixed Cross-Channel Correlator)**.**

$$G^{\text{gf}}_{\text{cross}}(t) = 8 \sum_{x,\mu} \text{Re}\big[b^{\text{gf}}_t(x,\mu)\, \bar{b}^{\text{gf}}_0(x,\mu)\big], \tag{15}$$

where $b^{\text{gf}}(x,\mu) = (U^{g^{\text{best}}}_{\mu}(x))_{01}$ is the $(0,1)$ element of the best-copy gauge-fixed link from P6-F3.

The ensemble average $\langle G^{\text{gf}}_{\text{cross}}(t)\rangle$ is the mean over configurations. Reproducibility under repeated measurement (deterministic arithmetic) is verified by D24. The gauge-fixing correction $\Delta^{\text{gf}}(t;c) = G^{\text{gf}}_{\text{cross}}(t;c) - G_{\text{cross}}(t;c)$ is generically non-zero since G_{cross} is gauge dependent.

4.2 Weak-Coupling Identification

For $A = A^a \tau^a$ traceless antihermitian, the SU(2) Cayley formula gives $\exp(igA) = \cos(g|A|/2)\mathbb{I} + (2i/|A|)\sin(g|A|/2)A$, so

$$b = (\exp(igA))_{01} = (ig/\sqrt{2})\, W\big[1 - g^2|A|^2/24 + O(g^4)\big], \tag{16}$$

with $W \equiv (A^1 - iA^2)/\sqrt{2}$. Because $A^2 = (|A|^2/4)\mathbb{I}$ for traceless antihermitian 2×2 A, there is no $O(g^2)$ correction to the off-diagonal element:

$$b = (ig/\sqrt{2})\, W + O(g^3). \tag{17}$$

This SU(2) improvement tightens the Theorem 2 error from $O(g)$ to $O(g^2)$ relative. Conjugating $(ig \to -ig)$,

$$\text{Re}[b_t \bar{b}_0] = \tfrac{g^2}{4}(A^1_t A^1_0 + A^2_t A^2_0) + O(g^4), \tag{18}$$

the positive sign arising from $(ig)(-ig) = +g^2$ (the locked Paper 5 Q5.3 sign). Hence

$$G^{\text{gf}}_{\text{cross}}(t) = 2g^2 \sum_{x,\mu}(A^1_t A^1_0 + A^2_t A^2_0) + O(g^4). \tag{19}$$

4.3 Theorem 2

Theorem 2 (Gauge-Fixed $W^\pm$ Propagator; paper-local)**.** *In Landau gauge (Lemma 4), on best-copy gauge-fixed configurations (Proposition 7, P6-F3), the zero-spatial-momentum $W^\pm$ temporal propagator satisfies*

$$D_{W^\pm}(t) = \frac{G^{\mathrm{gf}}_{\mathrm{cross}}(t)}{8Vg^2} + \delta_{\mathrm{pert}} + \delta_{\mathrm{Gribov}} + \delta_{\mathrm{stat}}, \tag{20}$$

where

$$D_{W^\pm}(t) \equiv \frac{1}{4V} \sum_{x,\mu} (\langle A^1_\mu(x,t) A^1_\mu(x,0)\rangle + \langle A^2_\mu(x,t) A^2_\mu(x,0)\rangle). \tag{21}$$

The leading-order identification $D_{W^\pm}(t) = G^{\mathrm{gf}}_{\mathrm{cross}}(t)/(8Vg^2) + O(g^2)D^{(0)}$ holds with $O(g^2)$ relative error (not $O(g)$), by the SU(2) *group identity of §4.2. Using $g^2 = (16/3)(1 - \langle P\rangle) = 8/\beta_{\mathrm{eff}} + O(\beta_{\mathrm{eff}}^{-2})$,*

$$D_{W^\pm}(t) = G^{\mathrm{gf}}_{\mathrm{cross}}(t)\, \frac{\beta_{\mathrm{eff}}}{64V} + O(\beta_{\mathrm{eff}}^{-1}). \tag{22}$$

L/R caveat: the embedding does not, by itself, distinguish $\mathrm{SU}(2)_L$ *from* $\mathrm{SU}(2)_R$*; this requires external fermionic input.*

Proof. Step 1 (gauge-fixing validity). By Lemma 4, P6-F1/F2 converge to $\varepsilon_{\mathrm{gauge}} = 10^{-14}$ (D19); the per-site condition holds (D20); by Proposition 7 and P6-F3 the best copy is selected and stable (D23), so $G^{\mathrm{gf}}_{\mathrm{cross}}$ is well defined and reproducible. *Step 2 (expansion).* From (17)–(18) the relative perturbative error is $O(g^2)$. *Step 3 (identification).* From (19) and (21), $8Vg^2 D_{W^\pm}(t) = 2g^2 \sum_{x,\mu}(A^1A^1 + A^2A^2)$ exactly, so $G^{\mathrm{gf}}_{\mathrm{cross}}(t) - 8Vg^2 D_{W^\pm}(t) = O(g^4)$ and the stated identification follows. *Step 4 (coupling).* Using $g^2 = (16/3)(1 - \langle P\rangle)$ introduces $O(g^2)$ relative error, consistent with Step 2. *Step 5 (remainders).* δ_{Gribov} is bounded by Proposition 7 (D22–D23); $\delta_{\mathrm{stat}} \sim 1/\sqrt{N_{\mathrm{cfg}}}$. □

Remark (Provenance)*.* The Paper 5 §8.3 identification was perturbative only, not gauge-fixed, without Gribov analysis, and unverified. Theorem 2 supplies all four, plus the $O(g^2)$ (rather than $O(g)$) error bound from the SU(2) identity. It is a new result of Paper 6.

4.4 Error Budget

With $D^{(0)}(t) \equiv G^{\mathrm{gf}}_{\mathrm{cross}}(t)/(8Vg^2)$, the three independent terms are: the perturbative correction $|\delta_{\mathrm{pert}}|/D^{(0)} \le C_P g^2 \approx 8C_P/\beta_{\mathrm{eff}}$ (so $\sim 40\% C_P$ at $\beta_{\mathrm{eff}} = 20$, $O(1)$ for $\beta_{\mathrm{eff}} \lesssim 8$); the Gribov systematic $\delta_{\mathrm{Gribov}}(t) = \Delta {G^{\mathrm{gf}}_{\mathrm{cross}}}^{\max}(t)/(8Vg^2)$ from D22–D23; and the statistical uncertainty $\sigma_{\mathrm{stat}}[D_{W^\pm}(t)] = \sigma[G^{\mathrm{gf}}_{\mathrm{cross}}(t)]/(8Vg^2\sqrt{N_{\mathrm{cfg}}})$. Combined in quadrature,

$$[\delta D_{W^\pm}(t)]^2 = \delta^2_{\mathrm{pert}} + \delta^2_{\mathrm{Gribov}} + \delta^2_{\mathrm{stat}}. \tag{23}$$

4.5 Corollary 6: Sign Structure of $G^{\mathrm{gf}}_{\mathrm{cross}}$

Corollary 6 (Sign Structure of $G^{\mathrm{gf}}_{\mathrm{cross}}$; series-wide)**.**

(i) At leading order $O(g^2)$, the ensemble average satisfies $\langle G^{\mathrm{gf}}_{\mathrm{cross}}(t)\rangle \ge 0$ for all $t \ge 0$. This is an ensemble-average statement, not pointwise: individual configurations may be negative at finite volume.

(ii) At non-perturbative level, $\langle G^{\mathrm{gf}}_{\mathrm{cross}}(t)\rangle$ is not sign-definite once $O(g^4)$ and genuine non-perturbative contributions dominate. A threshold β_c is defined as the coupling below which the average turns negative for some t.

(iii) The sign transition at β_c is a diagnostic for the onset of non-perturbative dynamics in the $W^{\pm}$ channel; its β_{eff}- and N-dependence are the empirical subject of Section 7.

Proof. *(i)* From (19), $\langle G^{\text{gf}}_{\text{cross}}(t)\rangle = 2g^2 \sum_{x,\mu}\langle A^1_t A^1_0 + A^2_t A^2_0\rangle + O(g^4)$; at strict $O(g^2)$ the right-hand side is the *free* lattice gauge-field two-point function, which is reflection positive, so $\langle G^{\text{gf}}_{\text{cross}}(t)\rangle \geq 0$ for $t \geq 0$. (This is a free-theory statement: the interacting Landau-gauge gluon propagator is *not* reflection positive, which is precisely the non-perturbative sign violation of part (ii) and Section 7.) *(ii)* By example at $\beta_{\text{eff}} = 2.0$, non-perturbative corrections are $O(1)$ and negative-valued configurations are present. □

Remark (Correction note). The original task specification stated $G^{\text{gf}}_{\text{cross}} \geq 0$ pointwise; this is too strong at finite volume. The correct statement is the ensemble average of part (i).

4.6 Checks D24 and D25

D24 (reproducibility) verifies $\max_{t,c} |G_1(t;c) - G_2(t;c)| < \varepsilon_{D24} = 10^{-12}$; a floating-point-floor result ($\sim 10^{-14}$) confirms purely deterministic arithmetic. D25 (weak-coupling consistency) at $\beta_{\text{eff}} \in \{12, 20\}$ requires (a) ensemble-average positivity, (b) cosh-like (periodic) behaviour of $D_{W^{\pm}}(t)$, symmetric about $t = N_t/2$, within 2σ, (c) consistency at $\beta_{\text{eff}} = 20$ with the near-massless free-mode form: the zero-momentum correlator is there dominated by the free mode, so a stable $m_{\text{eff}} > 0$ plateau is a non-perturbative (not a weak-coupling) expectation and is not required here, and (d) coupling consistency $|D^{\text{meas}}_{W} - D_{W^{\pm}}| < (16/\beta^2_{\text{eff}})D_{W^{\pm}}$. The $\beta_{\text{eff}} = 20$ component is conditional on the extended ensemble (Section 5).

5 Ensemble Extension and Volume Scaling

The base ensembles ($N = 6$, $N_{\text{cfg}} = 20$) carried two limitations: a $\sim 22\%$ statistical uncertainty and an unremovable finite-volume systematic from a single lattice size. Both are eliminated here.

5.1 Rationale and Specifications

Raising N_{cfg} from 20 to 100 reduces the fractional statistical error by $\sqrt{5} \approx 2.2$, below 10% for $\beta_{\text{eff}} \geq 5$. Extending to $N \in \{8, 12, 16\}$ exposes the $1/N^2$ finite-volume scaling and enables the thermodynamic extrapolation required for comparison with the SU(2) data of [3]. The base ensembles ($N = 6$, $\beta_{\text{eff}} \in \{5, 6, 9, 12\}$) are reprocessed with $N_{\text{cfg}} = 100$. Three groups of new ensembles are generated (Wilson plaquette action, heatbath plus overrelaxation, periodic boundary conditions, ≥ 500 thermalisation sweeps, ≥ 10 sweeps between measurements):

Group	Coverage	Count
E1 (strong coupling)	$N \in \{6, 8, 12, 16\}$, $\beta_{\text{eff}} \in \{2.0, 2.5, 3.0, 3.5\}$	16
E2 (perturbative)	$N \in \{6, 8, 12, 16\}$, $\beta_{\text{eff}} = 20.0$	4
E3 (FV scaling)	$N \in \{8, 12, 16\}$, $\beta_{\text{eff}} \in \{5, 6, 9, 12\}$	12

giving 32 new ensembles, all $N_{\text{cfg}} = 100$. Group E2 discharges FLAG-7 from Section 4 (the $N = 6$, $\beta_{\text{eff}} = 20$ ensemble required by D25). Algorithm selection follows Lemma 4: P6-F1 at $\beta_{\text{eff}} \leq 9$ or $N \leq 8$, P6-F2 at $\beta_{\text{eff}} \geq 9$ or $N \geq 12$.

5.2 Autocorrelation and Jackknife Errors

The integrated autocorrelation time $\tau_{\text{int}} = \frac{1}{2} + \sum_{\tau \geq 1} \rho(\tau)$ is estimated on $G^{\text{gf}}_{\text{cross}}(t = 1)$ with the Madras–Sokal window. Configurations are independent when $n_{\text{sep}} \geq 2\tau_{\text{int}}$; the provisional $n_{\text{sep}} = 10$ is increased where $\tau_{\text{int}} > 5$. All errors in Sections 5 to 7 are jackknife errors,

$$\sigma^2_{\text{JK}}(t) = \frac{N_{\text{cfg}} - 1}{N_{\text{cfg}}} \sum_c \big(\bar{G}_{-c}(t) - \bar{G}(t)\big)^2, \tag{24}$$

computed on configuration-level data binned into blocks of length $\geq 2\tau_{\text{int}}$ (block jackknife), so that residual autocorrelation is captured, $\sigma^2_{\text{JK}}/\sigma^2_{\text{naive}} \approx 2\tau_{\text{int}}$ for the binned estimator. A delete-one jackknife on unbinned, autocorrelated configurations would instead return $\approx \sigma^2_{\text{naive}}$ and *underestimate* the error; the bin length is increased until σ_{JK} plateaus.

Check D26. With coefficient of variation $\text{CV}(t{=}1) = \sigma_{\text{JK}}(t{=}1)/|\langle G^{\text{gf}}_{\text{cross}}(t{=}1)\rangle|$, the pass condition is $\text{CV} < 5\%$ for all ensembles plus running-mean stability. PARTIAL PASS is expected at strong coupling ($\beta_{\text{eff}} \leq 3.5$), where non-perturbative fluctuations inflate the variance.

5.3 Finite-Volume Scaling

The zero-momentum propagator on an N^3 spatial lattice is fitted with the phenomenological finite-volume ansatz

$$D_{W^\pm}(t; N) = D^\infty_{W^\pm}(t) + \frac{a(t)}{N^2} + O(N^{-4}), \tag{25}$$

with $a(t)$ a fit parameter. The $1/N^2$ form is *not* obtained from Euler–Maclaurin: for a smooth periodic summand on a full-period grid the Euler–Maclaurin boundary terms cancel and the residual error is super-algebraically small, so any power-law correction must originate elsewhere — in the long-distance (near-massless / infrared) non-analyticity of the two-point function and in the treatment of the spatial zero mode. The $1/N^2$ leading power is therefore adopted empirically and validated by the fit. For a decoupling (massive, $m_{\text{IR}} > 0$) propagator the leading finite-volume correction is instead expected to be exponential, $\sim e^{-m_{\text{IR}}N}$; both the $1/N^2$ and an e^{-mN} form are fitted and the better-supported one retained, with the N^{-4} term monitored at small β_{eff}. The ansatz presumes $m_{\text{IR}}N \gg 1$ (expected for $\beta_{\text{eff}} \geq 5$). For each (t, β_{eff}), (25) is fitted across $N \in \{6, 8, 12, 16\}$ by weighted least squares ($\chi^2/\text{dof} = \chi^2/2$, target $\lesssim 1$ at $\beta_{\text{eff}} \geq 5$), yielding $D^\infty_{W^\pm}(t)$ with propagated jackknife uncertainty and a residual systematic $\delta_{\text{FV}}(t) = |a(t)|/256$.

5.4 Plaquette Coupling

The working formula (22) requires g^2 per ensemble, extracted as $g^2_{\text{meas}} = (16/3)(1 - \langle P\rangle)$ (Table B.1). At $\beta_{\text{eff}} = 20$ the deviation from $g^2_{\text{pert}} = 8/\beta_{\text{eff}}$ is a higher-order effect to be quantified from the measured $\langle P\rangle$; a deviation exceeding the one-loop estimate triggers substitution of the one-loop improved coupling. At strong coupling, large deviations are expected and reflect genuine non-perturbative physics, not failure.

6 Validation Against the SU(2) Landau-Gauge Gluon Propagator

We ask whether $D_{W^\pm}(t)$, extracted from the b-channel, agrees with independently measured SU(2) Landau-gauge gluon propagator data. This requires colour symmetry (§6.2), normalisation matching (§6.3), and the absence of dominant uncontrolled systematics.

6.1 The Comparison Target

Cucchieri–Mendes [3] measured the zero-momentum SU(2) Landau-gauge gluon propagator on large lattices (to $N = 128$, $\beta \in \{2.2, 2.4\}$), averaging over all three SU(2) colour components and all directions, and found $D(p^2{=}0)$ finite and non-zero with strong infrared suppression — the decoupling solution, $D(p^2) \sim 1/(p^2 + m_{\mathrm{IR}}^2)$. This is our SU(2) benchmark. Bogolubsky et al. [4] report the same decoupling behaviour but in SU(3) gluodynamics (lattices to $L = 96$ at $\beta = 5.70$, with a simulated-annealing approach to the global gauge-functional maximum); it is cited here only as a qualitative, gauge-group-independent confirmation of the decoupling scenario and a precedent for global-maximum gauge fixing, *not* as an SU(2) quantitative benchmark.

> ***[OPEN]*** *The matched-β_{eff} Part B comparison (§6.4) requires a genuine large-volume* SU(2) *dataset. A suitable reference is Bornyakov, Mitrjushkin and Müller-Preussker,* Phys. Rev. D ***81*** *(2010) 054503 (*SU(2) *Landau-gauge gluon propagator; continuum and infinite-volume limits, infrared mass scale). It must be added to the bibliography and used in place of [4] for any quantitative* SU(2) *comparison.*

6.2 SU(2) Colour Symmetry

Proposition 8 (Colour Symmetry of the Landau-Gauge Propagator)**.** *In* SU(2) *lattice gauge theory in Landau gauge,*

$$D^{a=1}(p^2) = D^{a=2}(p^2) = D^{a=3}(p^2) \equiv D(p^2). \tag{26}$$

Proof. The Wilson action is invariant under the global adjoint action $A^a_\mu \to R^{ab} A^b_\mu$, $R \in \mathrm{SO}(3)$. Since SO(3) acts transitively on $\{1, 2, 3\}$, for any a, b there is R with $R\hat{e}_a = \hat{e}_b$; the Landau condition and F_L are invariant, so $\langle (A^a_\mu)^2 \rangle = \langle (A^b_\mu)^2 \rangle$, giving (26). (Transitivity of SO(3) on the adjoint index is specific to SU(2).) □

Consequently $D_{W^\pm}(p^2) = D^{a=1}(p^2) = D(p^2)$: the b-channel propagator equals the three-colour average used by [3], so the comparison is like-for-like. The W^3/A^3 component (measured by G_{diag}) must equal $D_{W^\pm}$ by (26), providing an independent cross-check (S6-2, §6.6); no $\mathrm{SU}(2)_L/\mathrm{SU}(2)_R$ distinction is involved.

Status of this comparison. Equation (26) fixes how this section should be read: in the SU(2) theory the b-channel propagator $D_{W^\pm}$ *is* the standard Landau-gauge gluon propagator, not a distinct observable. The novelty of the b-channel construction is therefore *structural* — $D_{W^\pm}$ is definable, and cleanly isolated, within the $\mathrm{Cl}(0, 2)$ framework — rather than *informational*: it carries no physical content that the standard gluon propagator does not already express. Agreement with the reference data is consequently *expected* by colour symmetry, so this section is a validation and implementation cross-check of the $\mathrm{Cl}(0, 2)$ extraction, not a new physical determination. A genuinely new propagator would require breaking the colour symmetry of (26) — e.g. through the fermionic $\mathrm{SU}(2)_L/\mathrm{SU}(2)_R$ content of Section 8 — which is beyond the pure-gauge scope of this section.

6.3 Normalisation Matching

With $D_{W^\pm}(t)$ as in (21), the temporal DFT gives $\widetilde{D}_{W^\pm}(\omega{=}0) = \sum_t D_{W^\pm}(t)$. Converting to the [3] convention requires tracking four factors. (i) *Colour.* By (26), $D_{W^\pm} = (1/4V) \sum_{x,\mu} (\langle A^1 A^1 \rangle + \langle A^2 A^2 \rangle) = (1/2V) \sum_{x,\mu} \langle A^1 A^1 \rangle$ is a two-charged-colour *sum*, whereas [3] uses the three-colour *average* $D = \frac{1}{3} \sum_a D^a$; with $D^1 = D^2 = D^3$ this is a relative factor $\frac{1}{2}$ (our sum is twice their average). (ii) *Direction.* A factor $1/d$ converts our direction sum to their per-direction average. (iii) *Field convention.* The complex-field relation $C_{W^\pm} = 2D_{W^\pm}$ (Paper 5) contributes a further factor that must be confirmed against the Paper 5 definition. (iv) *DFT/volume.* The $1/N_t$

DFT normalisation, with V as fixed by the Papers 4–5 convention (spatial N^3 vs total N^4). Collecting these,

$$D_{\mathrm{CM}}(p^2{=}0) = \kappa_{\mathrm{CM}} \, \frac{\sum_t G^{\mathrm{gf}}_{\mathrm{cross}}(t)\, \beta_{\mathrm{eff}}/(64V)}{d\, N_t}, \tag{27}$$

where $\kappa_{\mathrm{CM}} = \frac{1}{2}\, c_{\mathrm{conv}}$ absorbs the colour factor $\frac{1}{2}$ of (i) together with the field-convention and zero-mode constants of (iii)–(iv).

> ***[OPEN]*** *κ_{CM} is not pinned in this draft. It must be derived from the verbatim [3] zero-mode definition $\hat{A}^a_\mu(p{=}0)$ and the inherited Papers 4–5 V convention before any quantitative comparison; the colour factor $\frac{1}{2}$ is established here, the remainder is deferred.*

The sanity check S6-3 verifies the *implemented* conversion by comparing the DFT of $D_{W^\pm}(t)$ to the direct zero-mode computation (27), requiring their ratio to equal the implemented $d\, N_t/\kappa_{\mathrm{CM}}$ within 10^{-10}.

6.4 Scope Limitation and Two-Part Structure

Since [3] works at $\beta \in \{2.2, 2.4\}$ (our Scope D, ungenerated), and our available ensembles have $\beta_{\mathrm{eff}} \geq 5$, the comparison splits. **Part A (available):** at $\beta_{\mathrm{eff}} \in \{5, 6, 9, 12, 20\}$, $D_{\mathrm{CM}}(p^2{=}0)$ is computed via (27) and its β_{eff}-dependence examined as an internal consistency check of Theorem 2. **Part B (deferred):** at $\beta_{\mathrm{eff}} \in \{2.0, 2.5, 3.0, 3.5\}$, the FV-extrapolated $D^{\infty}_{W^\pm}(\omega{=}0)$ at $N = 16$ is compared directly to the SU(2) data of [3] (see the §6.1 [OPEN] note on adding a dedicated large-volume SU(2) reference); this requires Scope D generation and is fully specified here for later execution.

6.5 β_{eff}-Dependence and the Onset of Suppression

At weak coupling, $G^{\mathrm{gf}}_{\mathrm{cross}} \propto g^2 \propto \beta_{\mathrm{eff}}^{-1}$, so that $D_{W^\pm} = G^{\mathrm{gf}}_{\mathrm{cross}}\, \beta_{\mathrm{eff}}/(64V)$ is β_{eff}-independent at leading order; summing timeslices (cf. §7.4),

$$\widetilde{D}_{W^\pm}(\omega{=}0) = (d/2)\, \widetilde{C}_{\mathrm{free}}(\omega{=}0) + O(\beta_{\mathrm{eff}}^{-1}) \equiv P_\infty + O(\beta_{\mathrm{eff}}^{-1}), \tag{28}$$

a β_{eff}-independent plateau at leading order, with a sub-leading slope that is power-law (not exponential) in $\beta_{\mathrm{eff}}^{-1}$. The [3, 4] infrared suppression is a non-perturbative effect visible only at $\beta_{\mathrm{eff}} \sim 2$–$4$, where $\widetilde{D}_{W^\pm}(\omega{=}0)$ deviates *downward* from the plateau P_∞. Part A ensembles ($\beta_{\mathrm{eff}} \geq 5$) sit on the plateau; the onset of this downward deviation as β_{eff} decreases is the signal sought. At $\beta_{\mathrm{eff}} = 20$, $D_{\mathrm{CM}}(p^2{=}0)$ should match the plateau (28); deviations beyond the Section 4 budget would indicate an implementation error.

6.6 Systematic Budget and Cross-Checks

The total systematic on $\widetilde{D}_{W^\pm}(\omega{=}0)$ combines perturbative, Gribov, finite-volume, and statistical terms in quadrature, paralleling (23). The dominant term is δ_{stat} or δ_{FV} at $\beta_{\mathrm{eff}} \geq 9$, δ_{pert} near $\beta_{\mathrm{eff}} = 5$, and δ_{Gribov} at $\beta_{\mathrm{eff}} \leq 3.5$ (limiting Part B). The colour-symmetry cross-check S6-2 requires the normalised difference $|D_{\mathrm{diag}}(\omega{=}0) - D_{W^\pm}(\omega{=}0)|/\sigma_\Delta < 3$, where σ_Δ is the jackknife error of the per-configuration difference $D_{\mathrm{diag}} - D_{W^\pm}$ (not of either term separately, since the two channels are strongly correlated on shared configurations), for all available ensembles; agreement validates simultaneously the b-channel extraction, the $\mathrm{Cl}(0,2)$ decomposition, and the absence of colour-asymmetric gauge-fixing bias. Future requirement F9 is specified: Part A is runnable on available ensembles (pending execution); Part B is specified pending Scope D.

7 Sign Structure and β_{eff} Characterisation

This section delivers Corollary 6 Part (iii): the empirical characterisation of the threshold β_c below which $\langle G^{\text{gf}}_{\text{cross}}(t)\rangle$ loses sign-definiteness, closing Open Question OQ2 of Paper 5. No new numbered results are introduced; the content is computational. The observable is $D_{W^\pm}(t)$ (Theorem 2); Observable 2 does not appear.

> *L/R caveat: The embedding does not, by itself, distinguish* $\mathrm{SU}(2)_L$ *from* $\mathrm{SU}(2)_R$*; this requires external fermionic input (Section 8).*

7.1 Sign Analysis Protocol (P6-F6, P6-F7)

For each ensemble the jackknife estimator yields $\langle G^{\text{gf}}_{\text{cross}}(t)\rangle \pm \sigma_{\text{JK}}(t)$ for $t = 1, \dots, N_t/2$. Timeslices are classified, with a uniform 2σ threshold, as

$$\begin{aligned} \text{P (Positive):}\quad & \langle G^{\text{gf}}_{\text{cross}}(t)\rangle > +2\sigma_{\text{JK}}(t), \\ \text{N (Negative):}\quad & \langle G^{\text{gf}}_{\text{cross}}(t)\rangle < -2\sigma_{\text{JK}}(t), \\ \text{Z (Zero):}\quad & |\langle G^{\text{gf}}_{\text{cross}}(t)\rangle| \le 2\sigma_{\text{JK}}(t). \end{aligned} \tag{29}$$

Ensembles are typed Type-P (all P), Type-N (some N, none P), Type-M (both P and N), or Type-Z (all Z). Type-P is expected at weak coupling (Corollary 6(i)); Type-N/M is the non-perturbative signal; Type-Z marks the crossover. The classification is applied to both $G^{\text{gf}}_{\text{cross}}$ and $G^{\text{gf}}_{\text{diag}}$ (the latter defined via the a-element by P6-F6).

7.2 Threshold β_c Characterisation

$$\beta_c(N) = \sup\{\beta_{\text{eff}} : \text{ensemble type} \in \{N, M, Z\}\}. \tag{30}$$

If all available ensembles at N are Type-P, then $\beta_c(N) < 5.0$ pending Scope D — a strict upper bound, not a failure. When the transition is bracketed by adjacent couplings, $\beta_c(N)$ is estimated by linear interpolation in the Category-P fraction. If $\beta_c(N)$ is obtained as a float for $N \in \{6, 8, 12, 16\}$, the $1/N^2$ ansatz of Section 5 is applied,

$$\beta_c(N) = \beta_c^\infty + A/N^2 + O(N^{-4}), \qquad \chi^2/\text{dof} < 3. \tag{31}$$

If $\chi^2/\text{dof} \ge 3$ (FLAG-8), β_c^∞ is reported as a lower bound. If β_c^∞ is consistent with zero within $2\sigma_{\text{fit}}$, this is flagged as a significant secondary finding (sign-positivity in the thermodynamic limit), deferred pending Scope D.

7.3 Colour Symmetry Cross-Check

The diagonal correlator $G^{\text{gf}}_{\text{diag}}(t) = 8\sum_{x,\mu} \mathrm{Re}[a^{\text{gf}}_t \bar{a}^{\text{gf}}_0]$ is classified by the identical protocol. By colour symmetry (§6.2), the sign pattern of $G^{\text{gf}}_{\text{diag}}$ should agree ensemble-by-ensemble with $G^{\text{gf}}_{\text{cross}}$ at weak coupling; discrepancies are reported without suppression as inline flags (candidate causes: lattice artefacts, gauge-fixing residuals, finite-N boundary effects).

7.4 $\widetilde{D}_{W^\pm}(\omega{=}0)$ Plateau and Onset Coupling (P6-F8)

With $\widetilde{D}_{W^\pm}(\omega{=}0) = \sum_t D_{W^\pm}(t) = \sum_t \langle G^{\text{gf}}_{\text{cross}}(t)\rangle \beta_{\text{eff}}/(64V) + O(\beta_{\text{eff}}^{-1})$, the weak-coupling expansion gives $D_{W^\pm}(t) = (d/2)\bar{C}_{\text{free}}(t) + O(\beta_{\text{eff}}^{-1})$ with $\bar{C}_{\text{free}}$ a β_{eff}-independent lattice-geometry constant. Hence

$$\widetilde{D}_{W^\pm}(\omega{=}0) = (d/2)\,\widetilde{C}_{\text{free}}(\omega{=}0) + O(\beta_{\text{eff}}^{-1}), \tag{32}$$

flat in β_{eff} at leading order — a consequence of Theorem 2, with the plateau value $P_\infty = (d/2)\widetilde{C}_{\text{free}}(\omega{=}0)$. The sub-leading slope is power-law in β_{eff}^{-1} (not exponential). The onset coupling is $\beta_{\text{onset}} = \min\{\beta_{\text{eff}} : |\widetilde{D}_{W^\pm}(\omega{=}0;\beta_{\text{eff}}) - P_\infty| < 2\sigma_{\widetilde{D}}\}$, expected to agree with $\beta_c(N)$ within one β_{eff} step. A downward deviation below P_∞ at small β_{eff} is the decoupling signature of [3, 4].

7.5 Check D27

D27 is a composite. **D27a (standard, runnable):** for $N \in \{6, 8, 12, 16\}$ and $\beta_{\text{eff}} \in \{9, 12, 20\}$, $\langle G^{\text{gf}}_{\text{cross}}(t)\rangle > 0$ for all t within $2\sigma_{\text{JK}}$ (Type-P), confirming Corollary 6(i); twelve sub-conditions, PARTIAL PASS on $N \in \{6, 8\}$ if Scope C is ungenerated; any Category-N timeslice is a FAIL requiring revisiting D19–D23. **D27b (inverted, deferred):** at $\beta_{\text{eff}} = 2.0$ (Scope D), $\langle G^{\text{gf}}_{\text{cross}}(t)\rangle < -2\sigma_{\text{JK}}$ for at least one timeslice per N (PASS = negative sign observed). Per QS5, a fully positive result at $\beta_{\text{eff}} = 2.0$ is not a code error but implies $\beta_c(N) < 2.0$, reported as PARTIAL PASS with the bound. Composite: D27a PASS with D27b DEFERRED gives D27 = PARTIAL PASS, the current expected status.

8 Fermionic Sector: Setup and the Cross-Projected Propagator

This section establishes the fermionic groundwork by implementing the Future Requirements F1–F4 of Paper 5 §7.3 (paper-local functions P6-F10–P6-F13). It does not perform the full non-perturbative fermionic test, which is Paper 7. It (a) selects a fermion formulation and implements the Dirac operator, (b) defines and constructs the cross-projected propagator $S_{\text{cross}} = P_+^{\text{EW}} S\, P_-^{\text{EW}}$ on the gauge-fixed backgrounds of Sections 2 and 3, and (c) verifies in the weak-coupling limit that $S_{\text{cross}} \neq 0$ for a doublet (D28) and $S_{\text{cross}} = 0$ for a singlet (D29).

> *L/R caveat (second occurrence): The embedding does not, by itself, distinguish* $\mathrm{SU}(2)_L$ *from* $\mathrm{SU}(2)_R$*; this requires external fermionic input.*

SL1 is *addressed* here through the doublet/singlet structure, but the caveat is not retired: the full distinction requires the non-perturbative test of Paper 7.

8.1 Fermion Formulation Selection (P6-F10)

The Nielsen–Ninomiya theorem [9, 10] forbids a local, Hermitian, doubler-free, chirally symmetric lattice Dirac operator, so any chirally-sensitive prediction requires a bypass. Two are standard. Domain wall fermions (DWF) [11, 12] introduce a fifth dimension of extent L_s; chiral modes localise on opposite walls and exact chirality is recovered as $L_s \to \infty$, with corrections $O(e^{-\alpha L_s})$. Overlap fermions [14] satisfy the Ginsparg–Wilson (GW) relation [13] exactly at finite a via $D_N = (1/a)(1 + \gamma_5\,\mathrm{sgn}(H_w))$, at higher cost. We adopt DWF for Section 8 and Paper 7: the weak-coupling test needs only approximate chirality (at $\beta_{\text{eff}} = 20$, $L_s = 8$, the residual mass $m_{\text{res}} \ll g^2$), and the DWF code carries forward to Paper 7. The singlet vanishing (Lemma 5) is formulation-independent. Locked parameters: $M_5 = 1.8$, $r = 1$, $L_s = 8$, $m_f = 0.01$.

8.2 Dirac Operator and the Ginsparg–Wilson Relation (P6-F11)

The 4D Wilson kernel with domain wall height M_5 is

$$\begin{aligned} D_W(M_5)_{xy,\alpha\beta,ij} &= (4r + M_5)\delta_{xy}\delta_{\alpha\beta}\delta_{ij} \\ &\quad - \tfrac{1}{2}\sum_\mu \big[(r - \gamma_\mu)_{\alpha\beta} U^{ij}_\mu(x)\delta_{y,x+\hat\mu} + (r + \gamma_\mu)_{\alpha\beta} U^{\dagger ij}_\mu(x - \hat\mu)\delta_{y,x-\hat\mu}\big], \end{aligned} \tag{33}$$

where $U^{ij}_\mu(x)$ are the gauge-fixed links acting on the isospin indices i, j for a doublet, and replaced by δ_{ij} (no gauge coupling) for a singlet. The 5D DWF operator $D_{\text{DWF}}[s, s']$ adds bulk hops $-P_R\delta_{s',s-1}$, $-P_L\delta_{s',s+1}$ and boundary mass terms $m_f P_L \delta_{s,0}\delta_{s',L_s-1}$, $m_f P_R \delta_{s,L_s-1}\delta_{s',0}$, with $P_{L/R} = (1 \mp \gamma_5)/2$. The physical propagator is the boundary-to-boundary element $S(x,y) = [D_{\mathrm{DWF}}^{-1}]_{s=0,s'=L_s-1}(x,y)$. At finite L_s,

$$\gamma_5 D_{\mathrm{eff}} + D_{\mathrm{eff}}\gamma_5 = a\, D_{\mathrm{eff}}\gamma_5 D_{\mathrm{eff}} + O(e^{-\alpha L_s}), \tag{34}$$

the GW relation, exact as $L_s \to \infty$. At $L_s = 8$, $\beta_{\mathrm{eff}} = 20$, the violation is negligible relative to the $O(g^2)$ signal of D28.

8.3 Nielsen–Ninomiya Bypass

The Nielsen–Ninomiya theorem [9] states that correct continuum limit, absence of doublers, and exact chiral symmetry cannot hold simultaneously. DWF sacrifices exact chirality at finite L_s ($\{\gamma_5, D_{\mathrm{DWF}}\} \neq 0$) and recovers it as $O(e^{-\alpha L_s})$, a controlled bypass. SL2 is thereby partially addressed: the weak-coupling test is an NN-bypass result, while full chiral claims at non-perturbative coupling require $L_s \to \infty$ control (Paper 7). Lemma 5 below is NN-independent.

8.4 CG Solver (P6-F12)

$S(x,y)$ is obtained by solving $D_{\mathrm{DWF}}\Psi = \eta$ via conjugate gradient on the normal equations $D^\dagger D\Psi = D^\dagger\eta$, with relative residual tolerance $\varepsilon_{\mathrm{CG}} = 10^{-10}$ and even-odd preconditioning. For D28/D29 the source is at $x_0 = (0,0,0,0)$, wall $s = 0$, Dirac index $\alpha = 0$; the doublet source sets isospin $i = 0$ and is gauge-coupled, while the singlet source carries no isospin index and propagates freely.

8.5 The Cross-Projected Propagator (P6-F13)

The electroweak projectors (Paper 5 §7.3) are $P_+^{\mathrm{EW}} = \mathrm{diag}(1,0) \otimes \mathbb{I}_4$ and $P_-^{\mathrm{EW}} = \mathrm{diag}(0,1) \otimes \mathbb{I}_4$, with $P_+^{\mathrm{EW}} + P_-^{\mathrm{EW}} = \mathbb{I}_8$, $(P_\pm^{\mathrm{EW}})^2 = P_\pm^{\mathrm{EW}}$, $P_+^{\mathrm{EW}} P_-^{\mathrm{EW}} = 0$.

Definition 6 (Cross-Projected Fermion Propagator)**.** For a doublet $\psi = (\psi_u, \psi_d)^\mathsf{T}$ with propagator $S(x,y)$,

$$S_{\mathrm{cross}}(x,y) := P_+^{\mathrm{EW}}\, S(x,y)\, P_-^{\mathrm{EW}} = S_{ud}(x,y), \tag{35}$$

the upper-to-lower isospin block — the channel sensitive to W^+ exchange at $O(g)$. The scalar norm is $|S_{\mathrm{cross}}|^2(x,y) = \sum_{\alpha\beta} |[S_{\mathrm{cross}}(x,y)]_{\alpha\beta}|^2$.

Note. The citation here is to Paper 5 §7.3; the outline reference to "Paper 3" is a bibliography-key discrepancy.

8.6 Lemma 5: Singlet Vanishing

Lemma 5 (Singlet Vanishing of S_{cross}; paper-local)**.** *For a fermion field ψ transforming as an* SU(2) *singlet, $S_{\mathrm{cross}}(x,y) = P_+^{\mathrm{EW}} S(x,y) P_-^{\mathrm{EW}} = 0$ identically, for all x, y, at any β_{eff}, any lattice spacing, and any Dirac operator.*

Proof. A singlet carries no SU(2) index and does not couple to U_μ, so its propagator is scalar in isospin space, $S(x,y) = s(x,y)\,\mathbb{I}_2 \otimes M_{\mathrm{Dirac}}$, isospin-blind (the same propagator in both isospin components). Since $P_\pm^{\mathrm{EW}}$ act on the 2-dimensional isospin factor,

$$P_+^{\mathrm{EW}}\,[s(x,y)\mathbb{I}_2 \otimes M]\,P_-^{\mathrm{EW}} = s(x,y)\,[P_+^{\mathrm{EW}} P_-^{\mathrm{EW}}] \otimes M = 0,$$

using $P_+^{\mathrm{EW}} P_-^{\mathrm{EW}} = 0$. □

The proof uses neither chirality nor the form of the Dirac operator; the result is universal. A non-zero singlet S_{cross} is therefore a definitive code diagnostic, motivating the inversion of D29.

8.7 Weak-Coupling Argument: Doublet Non-Vanishing

At $O(g)$ the W^+ field couples via $b = (U)_{01} = (ig/\sqrt{2})W^+_\mu$, so the doublet propagator acquires an off-diagonal block $S_{ud}(x, x_0) = \sum_{z,\mu} S_0^{uu}(x, z)[gW^+_\mu(z)\Gamma_\mu]S_0^{dd}(z, x_0) + O(g^2)$. Squaring and averaging, $\langle|S_{\text{cross}}|^2\rangle \sim g^2\, V \times$ (free convolution) > 0, strictly positive. At $\beta_{\text{eff}} = 20$, $g^2 = 0.2$ and the estimate is $\langle|S_{\text{cross}}|^2\rangle \sim O(10^2)$, far above the D28 threshold. The argument is weak-coupling only; the non-perturbative regime is Paper 7.

8.8 Checks D28 and D29

D28 (standard). For a doublet source at $\beta_{\text{eff}} = 20$, $N = 6$, over $N_{\text{cfg}} \geq 10$ gauge-fixed configurations, $(1/N_{\text{cfg}}) \sum_{\text{cfg}} \sum_{x\neq x_0} |S_{\text{cross}}(x, x_0)|^2 > \varepsilon_{\text{ferm}} = 10^{-4}$. PASS confirms the perturbative prediction (expected $O(10^2)$) — the first numerical evidence that the b-channel gateway (Proposition 5, Paper 5) is active in the fermionic sector.

D29 (inverted). For a singlet source on the same configurations, $(1/N_{\text{cfg}}) \sum_{\text{cfg}} \sum_{x\neq x_0} |S_{\text{cross}}|^2 < 10^{-10}$. PASS (inverted) confirms Lemma 5. Per QS5, a non-zero result is not new physics but a code error (singlet constructed with isospin coupling, projector misapplied, or indices swapped); the tolerance 10^{-10} sits four orders above the double-precision floor. The composite §8 fermionic check passes only if both D28 and D29 pass.

8.9 Scope Limit Status After Section 8

SL1 is addressed (fermionic input introduced; caveat continues through Sections 9 and 10 pending Paper 7). SL2 is partially addressed (DWF bypass at $L_s = 8$; full chiral claims deferred). SL3 and SL4 are unchanged; SL5 remains discharged.

9 Discussion

Paper 6 advances the programme from a formal identification to a measured propagator. We summarise the contribution through five conclusions and update the open questions of Paper 5.

Conclusions. **(C1)** The gauge-fixing programme (P6-F1–P6-F9) and the $G^{\text{gf}}_{\text{cross}}$ measurement are specified, and Theorem 2 is established, promoting the Paper 5 §8.3 formal identification to an error-budgeted quantity (numerical execution of checks D19–D26 pending). **(C2)** Gribov copy variation is quantified (Proposition 7), providing the first systematic uncertainty budget for the b-channel propagator. **(C3)** $D_{W^\pm}(t)$ is extracted and a comparison to the SU(2) data of [3] is set up ([4] enters only as a qualitative SU(3) decoupling cross-reference); Part A (consistency, $\beta_{\text{eff}} \geq 5$) is runnable and Part B (direct comparison, $\beta_{\text{eff}} \leq 3.5$) awaits Scope D. **(C4)** The sign structure of $\langle G^{\text{gf}}_{\text{cross}}\rangle$ across β_{eff} is resolved (Corollary 6, Section 7), answering OQ2. **(C5)** The fermionic sector is set up: S_{cross} is defined and the weak-coupling doublet/singlet prediction tested (D28/D29), with the full non-perturbative test deferred.

Updated open questions. OQ1 (Landau-gauge measurement) is addressed (Section 2–Section 6); Gribov–Zwanziger region analysis remains open. OQ2 (sign-definiteness) is characterised (Section 7); the continuum-limit extrapolation remains open. OQ3 (SU(3) extension) remains open but is now more tractable with the gauge-fixing infrastructure in place. OQ4 (chirality/fermion content): the weak-coupling test is completed (Section 8); the strong-coupling chirality test is Paper 7's primary target. OQ5–OQ7 (electroweak mixing, higher-spin representations, full spacetime) are unchanged and longer-range, save that the temporal links are now live (OQ7 partially advanced).

10 Conclusions

Result 1 (Theorem 2). The Landau-gauge $W^{\pm}$ propagator $D_{W^{\pm}}(t)$ is obtained from $G^{\mathrm{gf}}_{\mathrm{cross}}$ via Theorem 2, promoting the Paper 5 §8.3 formal identification to a fully specified, error-budgeted result (numerical evaluation pending the verification programme) with an explicit three-term error budget.

Result 2 (Proposition 7 and Corollary 6). Gribov copy variation is quantified, and the sign structure of $\langle G^{\mathrm{gf}}_{\mathrm{cross}}\rangle$ is characterised as a function of β_{eff} and volume, with a threshold β_c and a $1/N^2$ extrapolation.

Result 3 (Lemma 5; D28/D29). The cross-projected propagator S_{cross} is defined and constructed; it vanishes identically for an SU(2) singlet (Lemma 5), and the weak-coupling doublet prediction of Paper 5 §7.3 is verified perturbatively.

NOT statement. Paper 6 does not resolve the full Gribov problem, does not perform the non-perturbative fermionic test, does not extend to SU(3), does not introduce electroweak mixing or $U(1)_Y$, and does not establish an $\mathrm{SU}(2)_L/\mathrm{SU}(2)_R$ distinction. The embedding does not, by itself, distinguish $\mathrm{SU}(2)_L$ from $\mathrm{SU}(2)_R$; this requires external fermionic input.

A Updated Check Programme (D19–D29)

ID	§	Description	Type	Status
D19	2	F_L convergence $\Theta < 10^{-14}$	standard	pending exec.
D20	2	per-site Landau condition	standard	pending exec.
D21	2	gauge-fixed link traces (3 parts)	standard	pending exec.
D22	3	Gribov non-uniqueness $\Delta G_{\mathrm{cross}} > 10^{-6}$	inverted	pending exec.
D23	3	best-copy stability under re-fixing	standard	pending exec.
D24	4	$G^{\mathrm{gf}}_{\mathrm{cross}}$ reproducibility $< 10^{-12}$	standard	pending exec.
D25	4	weak-coupling consistency (4 parts)	standard	spec.; β_{eff}=20 pending
D26	5	statistical convergence, CV $< 5\%$	standard	pending exec.
D27a	7	sign positivity, weak coupling (Type-P)	standard	pending exec.
D27b	7	sign violation, strong coupling	inverted	deferred (Scope D)
D28	8	doublet $\lvert S_{\mathrm{cross}}\rvert^2 > 10^{-4}$	standard	pending exec.
D29	8	singlet $\lvert S_{\mathrm{cross}}\rvert^2 < 10^{-10}$	inverted	pending exec.

Status key. No D19–D29 result is executed in this draft: "pending exec." denotes a check that is fully specified and runnable on the relevant available ensembles but whose numerical outcome is not yet reported here; "deferred (Scope D)" denotes a check awaiting ensemble generation; "β_{eff}=20 pending" denotes a check whose perturbative component is conditional on the Group E2 extended ensemble (Section 5). Headline tables and verification summaries are populated when these are run.

Inverted checks (D22, D27b, D29) pass when the "unexpected" direction is observed; each carries an explicit justification of what a standard-direction result would mean physically (QS5). Any slope or sign anomaly is documented with a resolution before a check is closed. Checks D1–D18 (Papers 4–5) remain closed.

B Extended Ensemble and Gauge-Fixed Configuration Parameters

Table B.1 records the reprocessed Paper 4/5 ensembles (now gauge-fixed): β_{eff}, N, N_{cfg}, gauge-fixing algorithm, $\varepsilon_{\text{gauge}}$, Gribov copies per configuration, and the best-copy F_L value. Table B.2 records the new extended ensembles ($\beta_{\text{eff}} \in \{2.0, 2.5, 3.0, 3.5, 20.0\}$, $N \in \{8, 12, 16\}$, $N_{\text{cfg}} \geq 100$): thermalisation sweeps, separation, and acceptance. Table B.3 records the fermionic parameters: formulation (DWF), domain wall height $M_5 = 1.8$, Wilson parameter $r = 1$, $L_s = 8$, bare mass $m_f = 0.01$, $\varepsilon_{\text{CG}} = 10^{-10}$, and number of source points. Table B.4 records the script assignments for D19–D29. Software versions: SciPy 1.16.3, NumPy, and the FFT and sparse-linear-algebra dependencies introduced for Fourier-accelerated fixing and the Dirac solver.

Paper 7 entry point. The full non-perturbative fermionic test at strong coupling is Paper 7; the first available identifiers there are D30, Proposition 8, Corollary 7, Theorem 3, and Lemma 6.

References

[1] T. Firestone, *Paper 4 of the* $\mathbb{R}\otimes\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$ *series*: Cross-channel correlators in the $\mathrm{Cl}(0,2)$ embedding of lattice SU(2).

[2] T. Firestone, *Paper 5 of the* $\mathbb{R}\otimes\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$ *series*: The Two-Observable Decomposition in $\mathrm{Cl}(0,2) \cong \mathbb{H}$; §7.3 (conditional prediction) and §8.3 (formal identification).

[3] A. Cucchieri and T. Mendes, *Constraints on the infrared behavior of the gluon propagator in Yang–Mills theories*, Phys. Rev. Lett. **100** (2008) 241601.

[4] I. L. Bogolubsky, E.-M. Ilgenfritz, M. Müller-Preussker, and A. Sternbeck, *Lattice gluodynamics computation of Landau-gauge Green's functions in the deep infrared*, Phys. Lett. B **676** (2009) 69–73.

[5] J. E. Mandula and M. Ogilvie, *Efficient gauge fixing via overrelaxation*, Phys. Lett. B **248** (1990) 156–158.

[6] V. N. Gribov, *Quantization of non-Abelian gauge theories*, Nucl. Phys. B **139** (1978) 1–19.

[7] D. Zwanziger, *Local and renormalizable action from the Gribov horizon*, Nucl. Phys. B **323** (1989) 513–544.

[8] I. M. Singer, *Some remarks on the Gribov ambiguity*, Commun. Math. Phys. **60** (1978) 7–12.

[9] H. B. Nielsen and M. Ninomiya, *Absence of neutrinos on a lattice. I. Proof by homotopy theory*, Nucl. Phys. B **185** (1981) 20–40.

[10] H. B. Nielsen and M. Ninomiya, *Absence of neutrinos on a lattice. II. Intuitive topological proof*, Nucl. Phys. B **193** (1981) 173–194.

[11] D. B. Kaplan, *A method for simulating chiral fermions on the lattice*, Phys. Lett. B **288** (1992) 342–347.

[12] Y. Shamir, *Chiral fermions from lattice boundaries*, Nucl. Phys. B **406** (1993) 90–106.

[13] P. H. Ginsparg and K. G. Wilson, *A remnant of chiral symmetry on the lattice*, Phys. Rev. D **25** (1982) 2649.

[14] H. Neuberger, *Exactly massless quarks on the lattice*, Phys. Lett. B **417** (1998) 141–144.